QM Basic

Formula Sheet - Closed-book Written Exam Attachment

This sheet is cumulative. Each section identifies the lecture in which its material is first introduced. Definitions, conditions, units, and interpretation remain examinable even when a formula is provided.

Notation used throughout

- Ω: sample space; A, B: events; A^c: complement.
- X, Y: random variables; x_i, y_i: observed values for row i.
- n: sample size; N: population size.
- $\mu = E[X]$, $\sigma^2 = \mathrm{Var}(X)$, and p: population parameters.
- $\bar{X}, S^2, \widehat{P}$: estimators; $\bar{x}, s^2, \hat{p}$: realised estimates.
- α: significance level; df: degrees of freedom.

Unless stated otherwise, tests on this sheet are **two-sided**.

Lecture 1 - Data summaries and units

For a category with count n_A among n observations,

$$\text{proportion} = \frac{n_A}{n}, \qquad \text{percentage} = 100\frac{n_A}{n}\%.$$

For values $x_1, \ldots, x_n$,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \text{range} = \max_i x_i - \min_i x_i.$$

A change from $a\%$ to $b\%$ is $b - a$ **percentage points**. Its relative percentage change is

$$100\frac{b-a}{a}\% \quad (a \neq 0).$$

Keep units throughout: a slope or rate has **outcome units per predictor unit**.

Lecture 2 - Probability foundations

$$0 \leq P(A) \leq 1, \qquad P(\Omega) = 1, \qquad P(A^c) = 1 - P(A).$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If $A \cap B = \varnothing$, then

$$P(A \cup B) = P(A) + P(B).$$

For a finite sample space with equally likely outcomes,

$$P(A) = \frac{|A|}{|\Omega|}.$$

A random variable is a function $X : \Omega \to \mathbb{R}$. An event is a subset of Ω, not a numerical variable.

Lecture 3 - PMFs, CDFs, and expectation

For a discrete random variable with support $\mathcal{X}$,

$$p_X(x) = P(X = x), \qquad p_X(x) \geq 0, \qquad \sum_{x \in \mathcal{X}} p_X(x) = 1.$$

The cumulative distribution function (CDF) is

$$F_X(a) = P(X \leq a) = \sum_{x \leq a} p_X(x).$$

For any CDF,

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

Expectation and transformed expectation are

$$E[X] = \sum_{x \in \mathcal{X}} x\, p_X(x), \qquad E[g(X)] = \sum_{x \in \mathcal{X}} g(x) p_X(x).$$

If $X \sim \text{Bernoulli}(p)$,

$$P(X = 1) = p, \quad P(X = 0) = 1 - p, \quad E[X] = p.$$

Lecture 4 - Variance and conditioning

Let $\mu = E[X]$. Then

$$\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2] - \{E[X]\}^2, \qquad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

For $X \sim \text{Bernoulli}(p)$,

$$\mathrm{Var}(X) = p(1-p).$$

The usual sample variance and SD are

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

For $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \cap B) = P(A \mid B)P(B).$$

If $B_1, \ldots, B_k$ partition Ω,

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i).$$

Bayes' rule is

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{k} P(A \mid B_j)P(B_j)}.$$

Lecture 5 - Joint PMFs and correlation

For discrete X, Y,

$$p_{X,Y}(x,y) = P(X = x, Y = y).$$

Marginal and conditional PMFs are

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}.$$

Discrete X and Y are independent when, for every x, y,

$$p_{X,Y}(x,y) = p_X(x)p_Y(y).$$

For independent events A, B with $P(B) > 0$, $P(A \mid B) = P(A)$.

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y].$$

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\mathrm{SD}(X)\,\mathrm{SD}(Y)}, \qquad -1 \leq \rho_{XY} \leq 1.$$

Sample covariance and correlation are

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}), \qquad r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

Independence implies zero covariance when the relevant expectations exist; zero covariance does not generally imply independence.

Lecture 6 - Continuous distributions

For a continuous random variable with PDF f_X,

$$f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1, \qquad P(X = x) = 0.$$

$$P(a < X \leq b) = \int_a^b f_X(x)\,dx = F_X(b) - F_X(a).$$

If $X \sim U(a, b)$,

$$f_X(x) = \frac{1}{b-a} \quad (a \leq x \leq b), \qquad P(c < X < d) = \frac{d-c}{b-a},$$

$$E[X] = \frac{a+b}{2}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1).$$

Approximately 68% lies within $\mu \pm \sigma$, and 95% within $\mu \pm 1.96\sigma$.

Lecture 7 - Samples and estimators

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \widehat{P} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{for a 0/1 variable.}$$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad df = n-1.$$

The degrees of freedom follow from the constraint

$$\sum_{i=1}^{n}(X_i - \bar{X}) = 0.$$

For sampling without replacement, the course's independence approximation uses the 10% condition:

$$n \leq 0.10N.$$

Random sampling supports generalisation to the sampling frame; random assignment supports causal treatment comparisons.

Lecture 8 - Sampling distributions and SEs

For independent, identically distributed observations with mean μ and variance σ^2,

$$E[\bar{X}] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \approx \frac{S}{\sqrt{n}}.$$

The CLT gives, for sufficiently large n,

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \mathrel{\dot\sim} N(0,1), \qquad \bar{X} \mathrel{\dot\sim} N\left(\mu, \frac{\sigma^2}{n}\right).$$

For an independent normal sample with unknown σ,

$$T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}.$$

For a sample proportion,

$$E[\widehat{P}] = p, \qquad \mathrm{SE}(\widehat{P}) = \sqrt{\frac{p(1-p)}{n}}.$$

Normal approximation working condition:

$$np \geq 10, \qquad n(1-p) \geq 10.$$

Multiplying n by c^2 divides the SE by c.

Lecture 9 - Confidence intervals and zero tests

General confidence-interval form:

$$\text{estimate} \ \pm \ \text{critical value} \times \text{SE}.$$

For a population mean with unknown σ,

$$\bar{X} \pm t_{1-\alpha/2,n-1}\frac{S}{\sqrt{n}}.$$

For 95% procedures, $z_{0.975} = 1.960$. Selected critical values are

df	5	10	20	30	60	∞
$t_{0.975,df}$	2.571	2.228	2.086	2.042	2.000	1.960

For a contrast $\delta = \mu - \mu_0$,

$$\widehat{\delta} = \bar{X} - \mu_0, \qquad T = \frac{\widehat{\delta} - 0}{S/\sqrt{n}}, \qquad df = n-1.$$

$$H_0: \delta = 0 \qquad \text{versus} \qquad H_1: \delta \neq 0.$$

The two-sided p-value is

$$p = P(|T_{df}| \geq |t_{obs}| \mid H_0)\,.$$

Decision at level α: reject H_0 if $p \leq \alpha$; otherwise do not reject. For the same two-sided t procedure, the level-α test rejects zero exactly when the $100(1-\alpha)\%$ interval excludes zero.

Lecture 10 - Tests, errors, and power

For $H_0: p = p_0$ against $H_1: p \neq p_0$, define $\delta_p = p - p_0$. The null test statistic is

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}, \qquad p\text{-value} = 2P(Z \geq |z_{obs}|).$$

Check $np_0 \geq 10$ and $n(1-p_0) \geq 10$ for the test. A large-sample proportion interval is

$$\hat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

The equivalent interval for $p - p_0$ subtracts p_0 from both endpoints.

$$\begin{aligned} \text{Type I error: reject } H_0 \text{ when } H_0 \text{ is true,} \\ P(\text{Type I error}) = \alpha, \\ \text{Type II error: do not reject } H_0 \text{ when a specified } H_1 \text{ is true,} \\ \text{Power} = P(\text{reject } H_0 \mid H_1 \text{ is true}). \end{aligned}$$

For a regression coefficient row,

$$t = \frac{b_j - 0}{\mathrm{SE}(b_j)}, \qquad b_j \pm t_{1-\alpha/2,df}\,\mathrm{SE}(b_j),$$

tests $H_0: \beta_j = 0$ against $H_1: \beta_j \neq 0$. In R output: `Estimate` is b_j, `Std. Error` is $\mathrm{SE}(b_j)$, `t value` is their ratio, and `Pr(>|t|)` is the two-sided p-value.

Always report the estimate and units, uncertainty, decision, population/model scope, practical importance, and causal limit.